Structuring Judgment, Stabilizing Responsibility:

Cognitive Friction in Procedural Texts

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Abstract

Procedural documents such as Standard Operating Procedures (SOPs) organize action and accountability in regulated work. Prior research on organizational responsibility has primarily examined how responsibility is anticipated or stabilized across social and evaluative processes, often treating procedural documents as coordination or standardization devices. Less attention has been paid to how procedural language itself structures the localized judgment demands that actors must resolve during execution. This study introduces cognitive friction as a document-centered framework for analyzing how procedural texts embed structurally produced interpretive demands within routine instructions.

Drawing on a corpus of 500 procedurally realistic sentences derived from publicly available regulatory and safety documentation, the analysis operationalizes cognitive friction as the co-occurrence of four structural features in procedural sentences: conditional branching, cross-references, exception structures, and discretionary qualifiers. Rule-based computational annotation was used to identify feature distributions and co-occurrence patterns across the corpus.

Results show that individual friction features are common across procedural genres, but structurally compounded judgment demands concentrate within a smaller subset of sentences where conditional structures co-occur with referential or discretionary cues. These high-friction configurations concentrate interpretive resolution within localized procedural units while maintaining prescriptive accountability. Across procedural sources, non-zero friction appears consistently, indicating that structurally embedded judgment demands are a general property of

procedural text architecture rather than an artifact of any single institutional genre.

By demonstrating that responsibility-relevant judgment demands can be operationalized and observed at the level of procedural sentence structure, the study conceptualizes cognitive friction as a document-level mechanism linking procedural language to the sociotechnical organization of accountability in regulated work, situating procedural texts within the broader process of responsibility production in regulated organizations as the procedural stage of responsibility production.

1 Introduction

Organizational procedures in regulated environments are governed by written documents: Standard Operating Procedures (SOPs), protocols, and compliance manuals (Berg, 1997; Smith, 2005). These procedural texts do not merely record organizational knowledge; they structure how actors interpret situations, sequence actions, and allocate responsibility under conditions of uncertainty (Suchman, 1987; Hutchins, 1995; Pentland & Feldman, 2005). Across domains such as healthcare, laboratory science, and industrial production, organizational action is routinely mediated through such documents, making procedural texts central infrastructures through which judgment and accountability are operationalized (Berg, 1997; Vaughan, 1996). Prior scholarship has largely treated procedural documents as carriers of standardized knowledge or coordination devices, leaving their role in structuring interpretive burden and stabilizing responsibility underexamined.

Despite the centrality of procedural documentation in organizational work, existing accounts of cognitive difficulty have primarily explained breakdowns in terms of individual misunderstanding, skill deficits, or cognitive overload (Simon, 1957; Reason, 1990). This perspective locates difficulty in the actor rather than in the structure of the procedural texts that guide action. As a result, how document structure shapes interpretive burden and stabilizes responsibility—through which one attribution becomes fixed among competing possibilities—remains insufficiently theorized in research on organizational cognition and accountability (Heimer, 2012; Vaughan, 1996).

This study introduces the concept of cognitive friction in procedural texts, defined as structural features of documents that increase the effort required to translate instructions into situated action and judgment, thereby shaping how responsibility is allocated during procedural work. Rather than viewing procedural difficulty as an individual cognitive limitation, cognitive friction conceptu-

alizes difficulty as emerging from document organization—such as conditional branching, deferred reference, and exception layering—that structures how actors navigate instructions and justify decisions. Cognitive friction thus refers not to task complexity or informational load per se, but to the interpretive demands produced by procedural text structure. In contrast to accounts that locate difficulty in actor cognition or skill, cognitive friction locates interpretive burden in document form, specifying how textual structure conditions interpretation during procedural action and stabilizes responsibility in procedural work.

Procedural texts not only guide action but also structure how responsibility becomes attributable in regulated organizational work. When procedural instructions embed contingent pathways, deferred criteria, or discretionary qualifiers, actors must resolve locally situated interpretations while remaining individually accountable for resulting action. Responsibility in such settings is therefore not solely stabilized through retrospective evaluation or institutional attribution processes but is partially pre-structured within the architecture of procedural language itself. By specifying how document structure organizes localized judgment demands, cognitive friction identifies a document-level mechanism through which procedural texts participate in the sociotechnical production of organizational accountability. Procedural texts thus form the procedural stage of responsibility production in regulated organizations.

The analysis draws on a corpus of organizational procedural texts used in regulated operational environments, including publicly available procedural guidance documents and laboratory safety manuals. By examining how friction features appear across procedural sentences and structure interpretation within instructional sequences, the study develops a document-centered account of how responsibility becomes stabilized through procedural action.

This study makes three contributions. First, it conceptualizes cognitive friction as a document-level property that structures interpretive burden in procedural work. Second, it provides a systematic corpus-based demonstration that structurally embedded judgment demands are observable within procedural sentence architecture, showing how friction features distribute and co-occur across procedurally realistic texts. Third, it links document structure to the stabilization of responsibility, showing how procedural texts function as sociotechnical infrastructures that organize judgment and accountability in organizational life (Bowker & Star, 1999; Orlikowski, 2007).

2 Conceptual Framing

Accounts of cognitive difficulty in organizational settings frequently treat breakdown, strain, and error as outcomes of human limitations (Reason, 1990; Norman, 1988). Whether framed in terms of information-processing load, task complexity, or executional failure, analytical attention is directed toward the capacities and actions of individuals. Within this orientation, formal documents are implicitly positioned as stable and neutral artifacts that transmit instruction, while responsibility for successful enactment rests primarily with the actor.

Such an analytical stance becomes unstable in highly regulated environments, where documentation does more than describe tasks. Procedural texts define the conditions under which interpretation becomes necessary, specify (or defer) criteria for decision-making, and establish the grounds upon which actions will later be evaluated. Judgment in these settings is not merely a moment of cognition but an event that is recorded, reviewed, and retrospectively scrutinized. Under these conditions, responsibility cannot be understood solely as an attribute of individual performance; it must also be examined as an effect of how judgment is structured, distributed, and stabilized within documentation practices (Suchman, 1987; Weick, 1995).

This study introduces the concept of cognitive friction to render structurally produced judgment demands analytically visible. Cognitive friction captures moments in which procedural language requires interpretation, reconciliation, or discretionary adjustment without fully specifying the criteria for doing so. These moments are consequential because the resulting judgments remain individually attributable, even when their necessity arises from textual design.

To observe these dynamics at a granular level, this study adopts the sentence as the minimal unit of analysis. Individual procedural sentences frequently encode conditional branching, obligation cues, cross-references, and exception handling within a single syntactic structure. By examining how such features co-occur, the framework renders visible how documentation practices organize judgment and, in doing so, participate in the production of responsibility.

The analysis is situated in the broader genre of regulated procedural documentation, particularly SOP-like instructional texts used in compliance and safety-critical environments. Such procedural documents combine prescriptive authority, auditability, and executional traceability within a single documentary form. Their sentences do not merely describe recommended action but define account-

able action under conditions of regulatory scrutiny. This artifact specificity is critical: cognitive friction in procedural texts emerges not simply from linguistic complexity but from the requirement that procedural language simultaneously prescribe action, anticipate exceptions, and sustain audit-relevant justification. The framework therefore treats procedural sentences as sociotechnical artifacts in which interpretive demands and responsibility conditions are co-encoded.

This genre orientation allows the analysis to extend beyond domain-specific SOPs to procedurally structured texts across regulatory and safety contexts. These arguments resonate with and extend prior document-centered and sociotechnical accounts that treat organizational texts as active mediators of coordinated action. Research in science and technology studies has long emphasized that inscriptions and formal representations do not merely reflect practice but participate in structuring it by stabilizing categories, distributing interpretive work, and rendering action accountable within institutional settings. The present analysis builds on this tradition by specifying how such structuring occurs at the level of procedural sentence architecture, where conditional, referential, and obligation structures organize localized judgment demands within rule-governed work.

Within organizational theory, documentation has similarly been examined as a mechanism of standardization and control, particularly in contexts of formalized compliance and auditable practice. However, prior work has tended to treat procedural texts primarily as carriers of prescribed action rather than as sites where interpretive resolution becomes structurally necessary. By introducing cognitive friction as an observable property of procedural language, this study shifts attention toward how documentation distributes decision criteria and allocates responsibility-relevant judgment to actors during execution.

Cognitive friction thus contributes to broader sociotechnical perspectives on accountability by linking textual design to the stabilization of attributable action. Rather than locating responsibility solely in retrospective evaluation or actor discretion, the framework highlights how procedural inscriptions pre-structure the conditions under which judgment becomes necessary and individually attributable. In doing so, it extends existing theories of documentation and organizational coordination by demonstrating how responsibility is partially organized within the architecture of procedural text itself.

3 Methods: Corpus Construction and Computational Annotation

This study examines how structurally embedded judgment demands are distributed across procedural documentation using a mixed corpus of synthetic and naturally occurring sentences that balances structural control with ecological validity. The inclusion of synthetic sentences does not aim to approximate natural prevalence but to analytically isolate structural configurations that are otherwise difficult to observe in uncontrolled organizational corpora.

The corpus consists of 500 procedural sentences sampled from three sources: synthetic SOP-like sentences designed to isolate common structural configurations of procedural language, publicly available procedural guidance documents, and laboratory safety and compliance manuals used in regulated operational environments. Of these, 100 sentences were synthetically constructed to isolate recurrent procedural structures, and 400 were derived from publicly accessible procedural and regulatory documentation. Together, these sources represent procedurally realistic documentation across regulatory and safety contexts while preserving analytic clarity of structural features.

Procedural text is segmented at the sentence level, consistent with the theoretical framing of cognitive friction as a property of procedural inscription embedded within localized syntactic structure. Each sentence is evaluated for the presence of five structural features associated with judgment demands (obligation cues, conditional branching, cross-references, exception structures, and discretionary qualifiers).

Obligation cues were excluded from the composite cognitive friction score because they function as the prescriptive baseline of procedural language rather than as structurally contingent sources of interpretive demand. While obligation signals mandatory action, cognitive friction arises when execution pathways, criteria, or judgment conditions remain partially specified. Feature detection is implemented using a rule-based computational annotation procedure to ensure reproducibility and consistency across the corpus.

Based on this annotation scheme, a composite cognitive friction score is computed for each sentence as the count of co-occurring friction features (conditional branching, cross-references, exception structures, and discretionary qualifiers). This operationalization captures localized interpretive demand as the within-sentence co-occurrence of friction features.

This corpus-based computational approach enables empirical observation of cognitive friction

as an observable property of procedural text structure. By combining constructed and naturally occurring sentences under a unified annotation framework, the analysis retains conceptual precision while extending beyond purely synthetic demonstration toward procedurally realistic documentation environments.

3.1 Annotation and Coding Procedure

Annotation of cognitive friction features was conducted at the sentence level using a rule-guided coding framework derived from the conceptual operationalization. Each sentence was evaluated for the presence of five previously defined structural features: obligation cues, conditional branching, cross-references, exception structures, and discretionary qualifiers. Coding criteria were specified to ensure interpretive consistency while preserving sensitivity to structural variation within procedural language.

Obligation cues were identified through modal or prescriptive constructions (e.g., shall, must, required to) indicating mandated action. Conditional branching was coded when sentences contained explicit conditional structures (e.g., if, when, unless) that introduced contingent procedural pathways. Cross-references were marked where sentences directed actors to external procedural elements, such as other sections, documents, or criteria sources. Exception structures were coded when language explicitly signaled deviation from default procedural flow (e.g., however, except, unless otherwise specified). Discretionary qualifiers were coded when sentences introduced actor-facing flexibility or justificatory latitude (e.g., may, as appropriate, if needed, based on judgment), thereby allocating interpretive resolution to the actor during execution.

Coding was conducted to capture structural presence rather than interpretive outcome. The objective was not to evaluate correctness of action or decision quality but to identify textual configurations that allocate interpretive resolution to actors during execution. This structural coding approach enables corpus-level aggregation while maintaining alignment with the theoretical definition of cognitive friction as a property of procedural inscription rather than individual cognition.

3.2 Synthetic Component and Structural Control

In addition to naturally occurring procedural sentences, the corpus includes a synthetic subset of procedurally realistic sentences deliberately constructed to isolate structural features associated

with judgment demands. The inclusion of synthetic sentences serves a methodological purpose rather than a substitution for empirical material. This component supports observation of structurally embedded judgment demands independent of domain-specific variation.

The sentence was selected as the minimal unit of analysis because moments of interpretive demand are frequently embedded within individual clauses rather than distributed evenly across documents. Sentence-level analysis captures the co-encoding of conditional, referential, exception, and discretionary cues within a single procedural unit, enabling comparable annotation across synthetic and naturally occurring texts.

The synthetic subset consists of 100 procedurally realistic sentences constructed to reflect common structural configurations of SOP-like instructional language while avoiding domain-specific content. Each sentence was annotated using the same rule-based feature scheme applied to the empirical corpus, enabling direct structural comparison across synthetic and naturally occurring procedural texts.

4 Results

4.1 Feature Distribution

The empirical corpus consists of 500 procedurally realistic sentences drawn from publicly available regulatory and procedural documentation, including WHO guidance, biosafety protocols, chemical safety procedures, and synthetic SOP-like constructions. All sentences were annotated using the rule-based computational procedure described above.

Across the corpus, conditional branching was the most prevalent cognitive friction feature, occurring in 33.8% of sentences, indicating that contingent execution pathways are routinely embedded at the sentence level of procedural documentation. Discretionary qualifiers appeared in 12.0% of sentences, while cross-references occurred in 11.8%. Exception structures appeared in 3.6% of sentences. These distributions indicate that procedurally realistic documentation frequently embeds contingent execution pathways and discretionary judgment conditions within individual sentences, while explicit exception handling remains less common.

Because cognitive friction is defined as the co-occurrence of structurally embedded judgment demands, feature combinations were examined at the sentence level. The majority of sentences

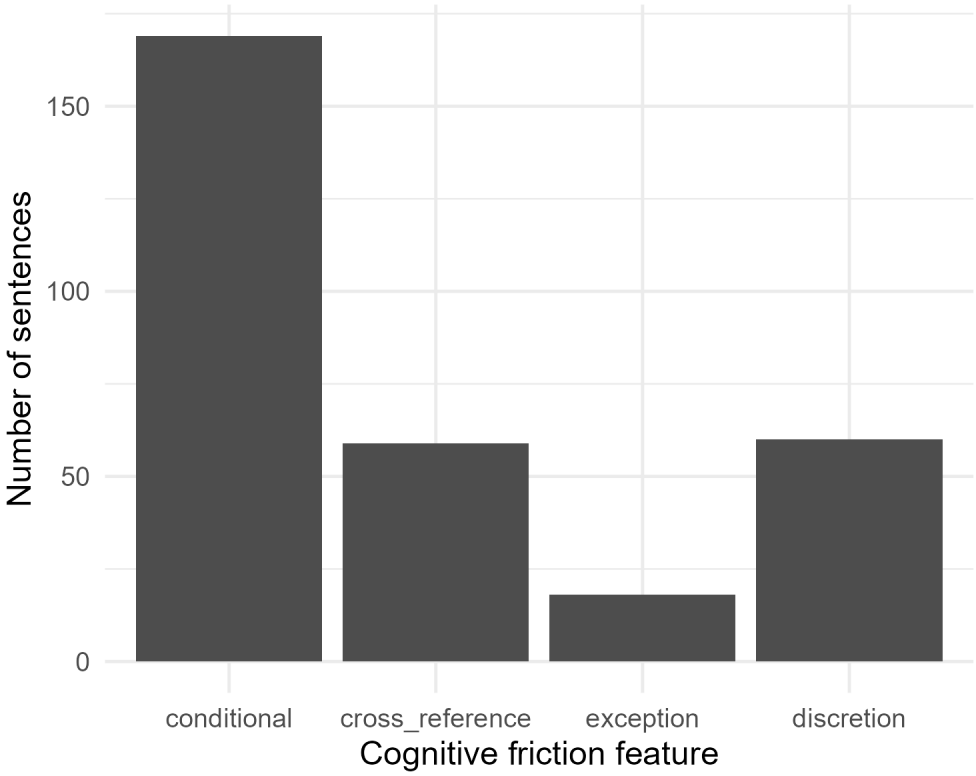

Figure 1: Frequency of cognitive friction features across procedural sentences in the empirical corpus (N = 500). Bars indicate the number of sentences containing each friction feature (conditional branching, cross-references, exception structures, discretionary qualifiers).

contained no friction features (50.2%), while 38.8% contained a single feature. Co-occurrence of two features occurred in 10.6% of sentences, and only 0.4% exhibited three simultaneous friction features. This distribution suggests that while individual friction cues are common, structurally compounded judgment demands are comparatively concentrated within a smaller subset of procedural sentences.

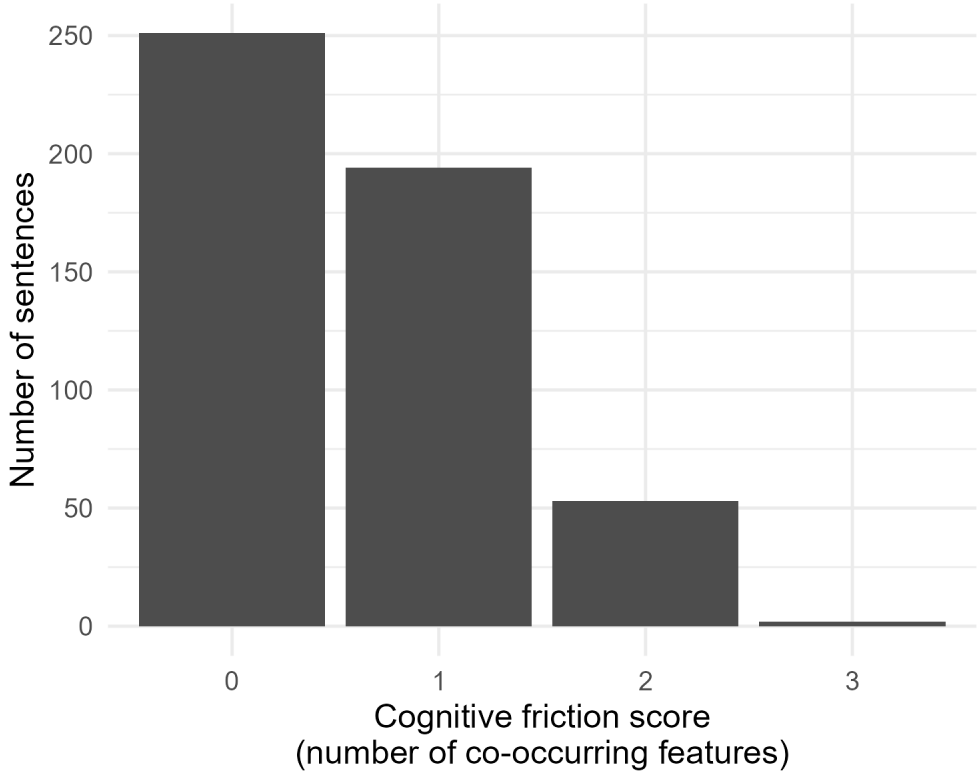

Figure 2: Distribution of composite cognitive friction scores across procedural sentences in the empirical corpus (N = 500). The friction score represents the co-occurrence count of structurally embedded judgment features (conditional branching, cross-references, exception structures, discretionary qualifiers) per sentence.

4.2 Composite Cognitive Friction Across Procedural Sources

To examine whether cognitive friction varies across procedural genres, mean composite friction scores were compared across source categories. Sentences derived from synthetic SOP-like constructions exhibited the highest average friction ($M = 0.95$), followed by chemical safety procedures ($M = 0.60$), WHO guidance texts ($M = 0.50$), and biosafety protocols ($M = 0.47$).

This gradient suggests that documents designed to simulate tightly specified procedural execution contexts concentrate structurally embedded judgment demands more densely than regulatory or guidance-oriented texts. In contrast, institutional guidance documents, while prescriptive, appear to distribute interpretive demands more diffusely across procedural context rather than within individual sentences.

Importantly, the presence of non-zero friction across all source categories indicates that cognitive friction is not limited to synthetic constructions but emerges across diverse forms of regulated procedural documentation. This supports the conceptual claim that structurally embedded judgment demands are a general property of procedural text architecture rather than an artifact of any single institutional genre.

4.3 Structural Concentration of High-Friction Sentences

Inspection of the highest-scoring sentences reveals that elevated cognitive friction arises when conditional branching co-occurs with cross-referential or discretionary structures within a single procedural statement. Such sentences simultaneously specify contingent execution conditions while deferring decision criteria to external references or localized judgment. This configuration produces concentrated interpretive demand because actors must resolve execution pathways under conditions of partial specification while remaining individually accountable for the resulting action.

Although such high-friction sentences are numerically rare, their structural configuration illustrates the mechanism through which procedural language can organize locally intensified judgment demands. These observations align with the conceptual definition of cognitive friction as structurally produced moments of interpretation embedded within procedural inscription.

4.4 Illustrative High-Friction Procedural Configurations

To illustrate how cognitive friction emerges in procedural text architecture, representative high-friction sentences were examined qualitatively. These examples demonstrate how co-occurring structural features concentrate interpretive demand within a single procedural statement.

Example 1 — Conditional + Cross-reference + Discretion *If reporting is delayed, the analyst shall notify the supervisor in accordance with SOP-QA-017 as deemed necessary.*

This sentence simultaneously introduces a contingent execution condition ("If reporting is delayed"), defers action specification to an external procedural reference ("in accordance with SOP-QA-017"), and embeds discretionary judgment ("as deemed necessary"). The actor must therefore determine both the applicability of the referenced SOP and the threshold for discretionary escalation, producing locally concentrated interpretive demand.

Example 2 — Conditional + Exception *When contamination is confirmed, the operator must continue processing except when safety criteria are compromised.*

Here, a conditional trigger ("When contamination is confirmed") is combined with an embedded exception structure ("except when safety criteria are compromised"). Execution requires the actor to reconcile competing procedural directives—continuation versus interruption—based on locally evaluated criteria. Responsibility for this reconciliation remains individually attributable despite originating from textual structure.

These configurations illustrate the mechanism by which cognitive friction arises: structurally embedded judgment demands co-occur within a single procedural unit, requiring actors to resolve contingent pathways under conditions of partial specification. Although such configurations are relatively infrequent in the corpus, their structural form demonstrates how procedural language can concentrate responsibility-relevant interpretation within localized textual architecture.

4.5 Co-occurrence Structure of Cognitive Friction

Beyond individual feature frequency, cognitive friction in procedural language emerges through the structural co-occurrence of multiple judgment-relevant cues within single sentences. Co-occurrence analysis examines how obligation, conditional, cross-reference, exception, and discretionary features combine to produce localized interpretive demand in procedural text.

5 Discussion

Across the empirical results, cognitive friction in procedural text emerges not from any single structural feature but from the co-occurrence of judgment-relevant features within individual sentences. While conditional branching, cross-references, discretionary qualifiers, and exception structures

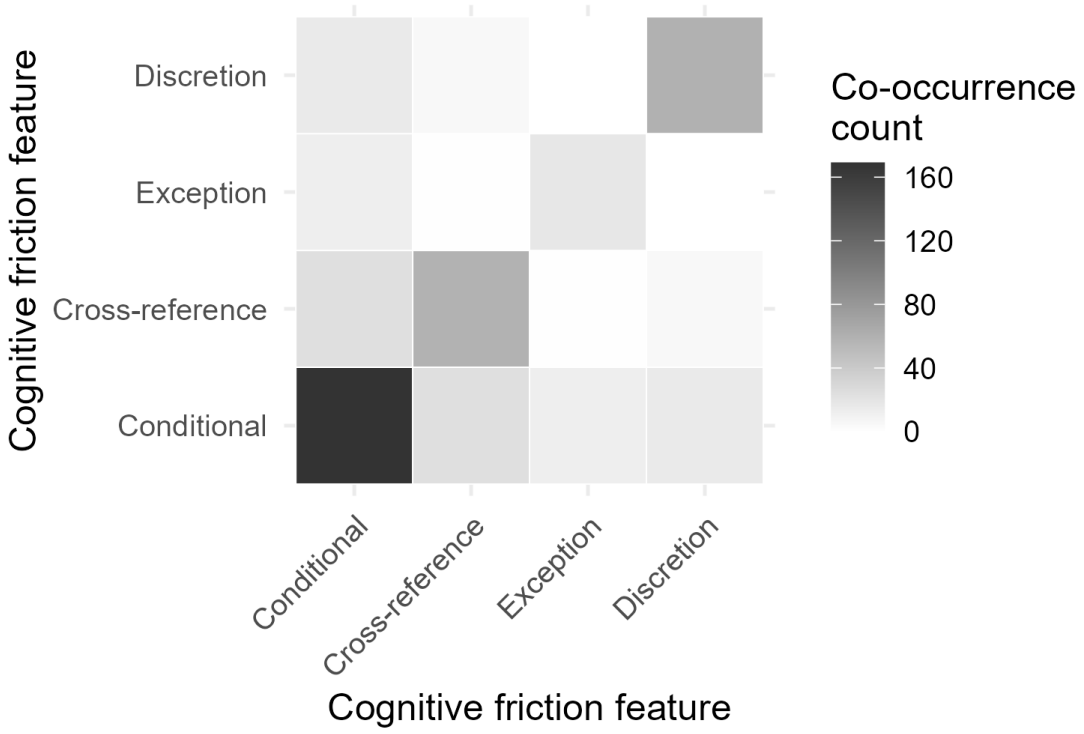

Figure 3: Co-occurrence structure of cognitive friction features across procedural sentences in the empirical corpus (N = 500). Cells indicate the number of sentences in which pairs of friction features (conditional branching, cross-references, exception structures, discretionary qualifiers) co-occur within the same procedural statement.

each appear across procedural documentation (Figure 1), compounded interpretive demand arises only in the smaller subset of sentences where these features interact (Figures 2–3). These configurations concentrate localized interpretive resolution at points where execution pathways remain partially specified yet individually accountable.

These empirical patterns clarify how responsibility-bearing judgment becomes structurally localized within procedural text. When friction features co-occur within individual sentences, interpretive resolution is concentrated at specific points of execution, thereby pre-structuring where accountable judgment must be exercised in procedural work.

This study introduces cognitive friction as a document-level property of procedural language that structures when responsibility-bearing judgment becomes necessary in regulated work. By operationalizing cognitive friction at the sentence level and examining its structural distribution and co-occurrence patterns, the analysis demonstrates that procedural sentences systematically embed configurations of conditional, referential, and discretionary structures that allocate localized interpretive resolution to actors during execution. These findings support the conceptualization of procedural documentation not merely as a carrier of standardized instruction but as a sociotechnical infrastructure that pre-structures the conditions under which accountable action must be produced (Tracey & Trethewey, 2005).

The distinction between cognitive friction scores and co-occurrence structure clarifies two complementary dimensions of procedural complexity: structural density and structural configuration. The friction score captures how many judgment-relevant features are embedded within a sentence, whereas co-occurrence analysis reveals how specific feature types combine to concentrate interpretive demand. Cognitive friction therefore arises not solely from feature quantity but from the interaction of structural elements within procedural inscription.

The synthetic component is used as an analytic control to isolate structural configurations of procedural language, not to estimate naturalistic prevalence. It supports the construct's structural observability by reducing domain and institutional confounds.

Although the analysis operates at the level of procedural language, its theoretical concern is not linguistic structure per se but the sociotechnical organization of responsibility in regulated practice. SOP sentences function as institutional inscriptions that prescribe action, encode contingencies, and establish audit-relevant justification (Latour, 1987; Power, 1997).

Within the broader research program on responsibility in regulated environments, cognitive friction identifies upstream textual conditions that generate judgment demands. Whereas anticipated accountability and responsibility convergence frameworks examine how responsibility is anticipated or stabilized across organizational processes, cognitive friction locates where responsibility-bearing judgment is structurally pre-configured within procedural documentation itself.

The framework therefore contributes a document-level mechanism linking procedural inscription to organizational accountability dynamics. More broadly, the findings suggest that responsibility in regulated organizational settings is not solely stabilized through retrospective evaluation or institutional attribution processes. It is partially generated within documentary infrastructures that pre-structure interpretive resolution during execution. This generative role of documentation is empirically reflected in the concentration of high-friction configurations observed across the procedural corpus. Procedural texts thereby function as accountability-bearing inscriptions that allocate judgment conditions prior to action, distributing responsibility through document architecture. This perspective extends sociotechnical and organizational theories of accountability by locating responsibility not only in evaluative regimes or actor performance but in the material organization of procedural inscription.

These findings have implications for the design and evaluation of procedural documentation in regulated organizations. If responsibility-bearing judgment is structurally concentrated within specific procedural configurations, documentation practices may inadvertently localize accountability at points where interpretive resolution is textually underspecified. Recognizing cognitive friction as a document-level property therefore suggests that procedural design itself participates in shaping downstream attribution of error, compliance, and responsibility in organizational evaluation processes.

Several limitations follow from the methodological design. The synthetic corpus isolates structural configurations but does not model full organizational accountability chains, attribution processes, or performance outcomes. Nor does the analysis estimate prevalence across real-world SOP corpora. Future empirical work may extend the framework to naturally occurring procedural documents and examine how cognitive friction relates to error investigation, audit processes, or responsibility attribution in practice. Nonetheless, by rendering structurally embedded judgment demands observable at the level of procedural sentence architecture, the present study establishes

cognitive friction as an analytically tractable construct for examining how responsibility is structurally organized within sociotechnical documentation systems.

Procedural sentences that embed structurally contingent execution pathways allocate localized interpretive resolution to actors while preserving individual accountability for resulting action. Cognitive friction thus identifies how procedural design distributes points of discretionary resolution within documentation, pre-structuring where responsibility-bearing judgment becomes necessary in regulated work.

This document-level account of responsibility structuring can be situated within a broader anticipatory process of accountability alignment in regulated organizational systems. Prior to action, actors often operate under convergent expectations regarding how their interpretations and executions will later be evaluated, justified, and attributed. Anticipated accountability convergence (AAC) conceptualizes this system-level alignment of accountability expectations across procedural, documentary, and evaluative infrastructures. From this perspective, cognitive friction specifies the localized textual mechanism through which anticipatory accountability structuring becomes embedded within procedural language itself. Procedural sentences that distribute interpretive resolution to executing actors therefore constitute sites where broader accountability convergence is operationalized at the document level. As illustrated in Figure 4, responsibility production in regulated sociotechnical systems can be understood as a multi-stage process spanning anticipatory alignment, procedural embedding, investigative stabilization, and representational mediation.

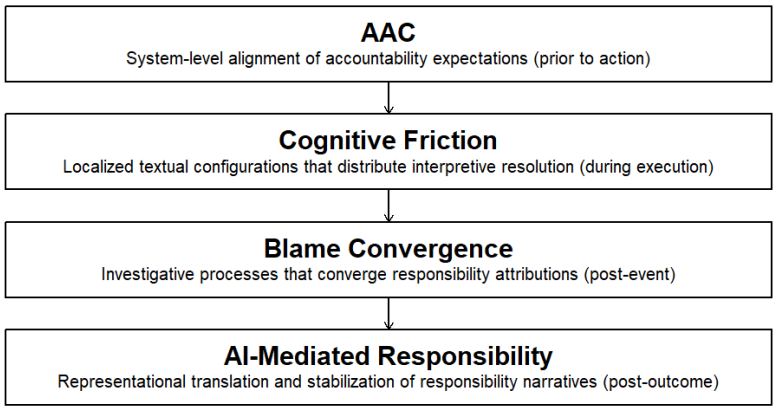

Figure 4: Responsibility production across stages of work in regulated sociotechnical systems. Responsibility is anticipated through system-level alignment of accountability expectations (anticipated accountability convergence; AAC), embedded within procedural language during execution (cognitive friction), stabilized through investigative processes following events (blame convergence), and subsequently mediated through representational translation and stabilization in sociotechnical evaluation systems (AI-mediated responsibility).

AAC operates at the level of system-wide alignment of accountability expectations across procedural and evaluative infrastructures prior to action. Cognitive friction, by contrast, specifies the localized textual configurations through which such anticipatory alignment becomes operationalized during execution.

6 Limitations and Scope Conditions

Several limitations delimit the scope of the present analysis and indicate directions for further empirical development. First, although the empirical corpus draws on multiple forms of publicly available procedural documentation—including laboratory guidance, biosafety protocols, chemical safety procedures, and synthetic SOP-like constructions—it remains restricted to highly formalized procedural genres. Such documents provide analytically clear instances of rule-governed inscription, but cognitive friction as a structural property of procedural language may vary across organizational domains with different documentation traditions, levels of formalization, or accountability regimes.

Future research may therefore examine how cognitive friction manifests in other procedural environments, including clinical, engineering, or administrative documentation systems. Cognitive friction thus provides a document-level lens on how procedural texts structure judgment and responsibility in regulated organizational work.

Second, the present study focuses on sentence-level structural features as minimal units of procedural inscription. This analytical choice enables identification of localized interpretive demands but does not capture higher-order document structures such as section organization, hierarchical nesting of procedures, or cross-document architectures that may also contribute to distributed judgment demands. Extending cognitive friction analysis to multi-sentence or document-level configurations would provide a more comprehensive account of how procedural texts structure interpretive work across organizational documentation systems.

Third, the coding framework identifies structural features associated with interpretive demand but does not measure cognitive effort or decision difficulty directly at the level of actors. Cognitive friction is conceptualized here as a property of procedural text rather than individual cognition; however, linking structural configurations to observed interpretive processes or performance outcomes would strengthen the empirical grounding of the framework. Future studies combining textual analysis with observational or experimental approaches could examine how structurally embedded judgment demands are experienced and resolved in situated practice.

Finally, the present analysis does not aim to establish causal relationships between procedural design and downstream responsibility attribution outcomes. While the framework specifies how procedural architecture allocates localized interpretive resolution to actors, accountability stabilization remains shaped by organizational evaluation processes, institutional norms, and interactional dynamics beyond the text itself. Cognitive friction therefore captures one upstream component of responsibility structuring within sociotechnical systems rather than a complete explanatory account.

These limitations nonetheless underscore the analytical value of treating procedural language as a site where responsibility-relevant judgment demands become structurally organized. By identifying observable textual configurations that allocate interpretive resolution within rule-governed work, the present framework provides a foundation for broader empirical and comparative investigations of how documentation practices participate in shaping the locus of accountable action

across organizational domains.

7 Future Research Directions

The cognitive friction framework opens several avenues for further empirical and theoretical development. One immediate direction involves extending analysis beyond sentence-level configurations to examine how multi-sentence structures and document architectures distribute interpretive demand across procedural systems. Procedural texts often embed decision criteria through hierarchical nesting, section-level dependencies, and cross-document linkages that cannot be fully captured at the level of isolated sentences. Investigating these higher-order textual structures would enable a more comprehensive account of how documentation organizes judgment across organizational knowledge infrastructures.

A second research direction concerns variation in cognitive friction across procedural domains and institutional contexts. Documentation practices differ substantially across industries in their levels of formalization, regulatory exposure, and accountability regimes. Comparative analysis across laboratory, clinical, engineering, or administrative procedural systems could reveal how domain-specific documentation conventions shape the distribution and intensity of structurally embedded judgment demands. Such comparisons would clarify whether cognitive friction reflects general properties of procedural language or domain-specific inscription practices.

Third, future work may examine how actors experience and resolve cognitive friction in situated practice. While the present study conceptualizes friction as a structural property of procedural text, its organizational consequences unfold through interaction with users interpreting and enacting procedural instructions. Observational or experimental studies could investigate how practitioners navigate conditional structures, deferred criteria, and cross-references during task execution, and how such navigation shapes decision trajectories and responsibility attribution. Linking textual configurations to observed interpretive processes would deepen the sociotechnical grounding of the framework.

Another promising direction involves integrating cognitive friction with broader theories of organizational routines and documentation infrastructures. Procedural texts form part of distributed organizational memory systems that coordinate action across time and actors. Examining

how friction-prone structures interact with routine dynamics, standardization processes, and audit regimes may illuminate how documentation both stabilizes and redistributes judgment within organizations. This perspective would position cognitive friction within ongoing theoretical debates on the performativity of organizational texts and the materialization of accountability structures.

Finally, computational approaches offer opportunities to scale cognitive friction analysis across large document corpora. Automated detection of structural features such as conditionality, cross-referencing, exception handling, and discretionary qualifiers could enable comparative mapping of friction distributions across procedural systems. Such analyses would allow examination of how documentation architectures evolve over time, respond to regulatory change, or vary across organizations. By combining qualitative interpretive analysis with computational text methods, future research could establish cognitive friction as a scalable analytical lens for studying documentation-centered coordination in complex sociotechnical environments.

8 Conclusion

This study demonstrates that responsibility-bearing judgment in regulated organizational work is partly structured within procedural sentence architecture and introduces cognitive friction as a document-level mechanism capturing this structural organization. By analyzing sentence-level configurations of structural features such as conditional branching, cross-references, exception structures, and discretionary qualifiers, the analysis demonstrates that interpretive demands are not solely products of individual cognition but are partially produced through the architecture of procedural language itself.

Situated within a broader research program on responsibility in regulated sociotechnical systems, cognitive friction specifies how responsibility-bearing judgment becomes structurally embedded within procedural documentation during execution. Procedural texts thereby function not only as instruments of coordination but as infrastructures of accountable action, shaping where interpretive resolution becomes necessary and where responsibility becomes attributable in organizational practice.

By foregrounding the procedural sentence as a minimal unit of inscription, this study identifies a document-level mechanism linking procedural design to the structuring of accountability. Future

research may extend this framework across domains and document architectures to examine how variations in procedural configuration influence the organization of judgment and responsibility in complex sociotechnical systems. Cognitive friction thus provides a document-level lens for analyzing how responsibility is materially organized within procedural systems, situating procedural texts within the broader process of responsibility production in regulated organizations as the procedural stage of responsibility production.

9 Code Availability

The code supporting this study is publicly available at:
`https://github.com/sorakim-phd/sop-cognitive-friction-pilot`

The repository contains the rule-based annotation scripts and analysis workflow used to identify sentence-level cognitive friction features in the 500-sentence procedural corpus analyzed in this study. The implementation operationalizes the structural feature definitions described in the Methods section (conditional branching, cross-references, exception structures, and discretionary qualifiers) and computes composite cognitive friction scores for each sentence.

The repository also includes an illustrative subset of procedural sentences and example annotation outputs to support computational transparency and reproducibility. Some example sentences are synthetic or lightly adapted for clarity, while the empirical corpus analyzed in the paper was derived from publicly accessible procedural and regulatory documentation.

The implementation is intended as a transparent research prototype demonstrating the computational observability of structurally embedded judgment demands in procedural text, rather than a predictive or production-level system.

10 Data Availability

The procedural sentence corpus analyzed in this study consists of 500 procedurally realistic sentences derived from publicly accessible regulatory and safety documentation, including WHO laboratory guidance, academic laboratory safety resources, and related procedural materials. Sentences were segmented, selected, and normalized at the procedural statement level for annotation.

Because the corpus includes sentences adapted from publicly available documents and synthetic constructions designed to approximate SOP-like procedural structure, only an illustrative subset and annotation outputs are provided in the public repository to support transparency and reproducibility. The full annotated corpus used in the analysis is available from the author upon reasonable request.

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